

AuthN = Authentication

AuthZ = Authorization

Network segmentation = control who/what can communicate with whom.

Host hardening = make each individual machine/container harder to compromise

LAB 4 · WEEKS 7–8**Access Control & Network Security***AuthN vs AuthZ, network segmentation and host hardening — Docker & Kubernetes*

Device A --> Device B --> Device C

Lab Learning Outcomes

At the end of this lab, you will be able to:

Device A

1. Distinguish and implement **authentication** (who you are) and **authorization** (what you may do).
2. Add a second factor with a **TOTP (MFA)** code and verify it.
3. Configure **network access control** and segmentation so services reach only what they must.
4. **Harden** a container image: non-root, minimal, dropped capabilities, read-only filesystem.
5. Scan an image for vulnerabilities and apply the principle of **least privilege** across compute, network and storage.

Course & Assessment Mapping

Item	Mapping
Course Learning Outcome	CLO2 — Construct secure cloud operations that safeguard data integrity
Lecture topics	Week 5 (Access Control) · Week 9 (Network Security patterns)
Value / skill clusters	VBE3 (Integrity) · SC8 (Integrated Problem-Solving)
Assessment	Lab report (screenshots + outputs + short answers) — contributes to the Lab Assignment

Lab Arrangement (2 Sessions over 2 Weeks)

Session	Week	Focus
Session A	Week 7	Authentication vs authorization, MFA, and RBAC enforcement (Tasks 1–3)
Session B	Week 8	Network segmentation, firewall rules and container hardening (Tasks 4–6), then the report

Note: Session A controls WHO gets in. Session B controls WHAT they can reach and reduces WHAT an intruder could exploit.

Technical Prerequisites

- A laptop with Docker and a terminal; **kind** and **kubectrl** installed.
- **oathtool** for TOTP (install via package manager) — free; or use any authenticator app.
- **Trivy** container scanner (or `docker scout`) — free.
- Internet only for first image downloads.

Security tip: Identity is the perimeter. Notice that almost every control in this lab ultimately asks the same two questions: are you who you claim, and are you allowed to do this?

Session A (Week 7) — Authentication & Authorization

Task 1 — Authentication: a Password-Protected Service

Run a web service behind **HTTP Basic authentication**. Only requests with **valid** credentials get in.

```
# Create a password file (user: student)
docker run --rm httpd:alpine htpasswd -nbB student 'P@ssw0rd!' > htpasswd.txt

# Serve a page that requires authentication Creates webpage using simple nginx
cat > default.conf <<'EOF'
server { listen 80;
    location / { auth_basic "Restricted";
        auth_basic_user_file /etc/nginx/.htpasswd;
        return 200 'Authenticated OK\n'; } }
EOF
docker run --rm -d --name authsvc -p 8080:80 \
    -v "$(pwd)/default.conf:/etc/nginx/conf.d/default.conf" \
    -v "$(pwd)/htpasswd.txt:/etc/nginx/.htpasswd" nginx

curl -s -o /dev/null -w 'no-creds: %{http_code}\n' http://localhost:8080 # 401 Unauthorized
curl -s -u student:'P@ssw0rd!' http://localhost:8080 # 200
Authenticated OK
```

Task 2 — Add a Second Factor (MFA / TOTP)

Passwords alone are **weak**. Generate a **time-based one-time password** and validate it, the same mechanism an authenticator app uses.

```
# Create a shared secret (base32) and generate the current 6-digit code
SECRET=$(head -c20 /dev/urandom | base32)
echo "Enrol this secret in an authenticator app: $SECRET"
oathtool --totp -b "$SECRET"

# Validate a code the user types (compare to the expected value)
read -p 'Enter the 6-digit code: ' CODE read "CODE?Enter the 6-digit code: "
[ "$CODE" = "$(oathtool --totp -b "$SECRET")" ] && echo 'MFA OK' || echo 'MFA FAILED'
```

FAST

```
SECRET=$(python3 -c 'import os,base64; print(base64.b32encode(os.urandom(20)).decode().rstrip("="))')
echo "Enrol this secret in an authenticator app: $SECRET"
oathtool --totp -b "$SECRET"
```

Note: MFA combines factors from different classes (something you know + something you have). It defeats the majority of credential attacks — the cheapest big security win.

Task 3 — Authorization: RBAC Roles

Authentication proves identity; **authorization** decides permissions. Create a cluster and compare a developer role with an admin role.

```
kind create cluster --name ccse-lab4
kubectl create namespace app
kubectl create serviceaccount dev -n app

# Developer may only read pods Permission
kubectl create role dev-role -n app --verb=get,list --resource=pods Read-only
kubectl create rolebinding dev-rb -n app --role=dev-role --serviceaccount=app:dev
```

```
SA=system:serviceaccount:app:dev
kubectl auth can-i list pods -n app --as=$SA # yes
kubectl auth can-i create deploy -n app --as=$SA # no
kubectl auth can-i delete pods -n app --as=$SA # no
```

CANNOT MODIFY

Note: End of Session A. Keep the 401/200 results, the MFA OK output, and the three can-i results. Stop the auth service (docker stop authsvc).

Session B (Week 8) — Network Security & Hardening

Task 4 — Network Segmentation (Three-Tier)

Separate a **frontend**, **backend** and **database** into isolated Docker networks so the frontend cannot reach the database directly — defence in depth for the network.

```
# Create two segmented networks
docker network create frontend-net
docker network create backend-net

# DB only on backend-net; app on both; web only on frontend-net
docker run -d --name db --network backend-net redis:alpine
docker run -d --name app --network backend-net nginx
docker network connect frontend-net app
docker run -d --name web --network frontend-net nginx

# web -> db should FAIL (not on the same network)
docker exec web sh -c 'apk add -q curl; curl -s -m 3 db:6379 || echo BLOCKED'
# app -> db should WORK (shared backend-net)
docker exec app sh -c 'apk add -q curl; nc -z -w3 db 6379 && echo REACHABLE'
docker exec app sh -c 'apt-get update -qq && apt-get install -y -qq netcat-openbsd && nc -z -w3 db 6379 && echo REACHABLE'
```

Security tip: The database is unreachable from the internet-facing tier. An attacker who compromises the web tier still cannot talk directly to the data — segmentation contains lateral movement.

Task 5 — Firewall Rules (Default-Deny)

Apply host-level firewall rules that permit only the ports you need. This mirrors cloud security groups.

```
# Inside a throwaway container with iptables, model default-deny + allow 443
docker run --rm --cap-add=NET_ADMIN alpine sh -c '\
  apk add -q iptables; \
  iptables -P INPUT DROP; \
  iptables -A INPUT -p tcp --dport 443 -j ACCEPT; \
  iptables -A INPUT -i lo -j ACCEPT; \
  iptables -L INPUT -n'
```

Note: Default policy DROP with a single explicit ACCEPT is the security-group model: nothing is allowed unless you permit it (least privilege for the network).

Task 6 — Container / Host Hardening

Reduce the attack surface. Build a minimal, non-root, capability-dropped, read-only container and scan it.

```
# A hardened run of a service
docker run -d --name hardened \
  --user 1000:1000 \           # non-root runs the service as a non-root user.
  --read-only \               # read-only root filesystem
  --cap-drop=ALL \            # drop all Linux capabilities removes extra Linux privileges.
```

Any normal non-zero UID, such as 1000, is a non-root user

```
--security-opt no-new-privileges \ prevents privilege escalation.  
--tmpfs /tmp \ gives the app a temporary writable /tmp area.  
nginxinc/nginx-unprivileged is an Nginx image designed to run without root  
  
docker inspect hardened --format 'User={{.Config.User}}  
ReadOnly={{.HostConfig.ReadonlyRootfs}}' verify that the hardening settings were actually applied.  
  
# Scan an image for known vulnerabilities  
docker run --rm aquasec/trivy image --severity HIGH,CRITICAL nginx:alpine | head -20
```

In your report, list three hardening measures you applied and the attack each one blunts.

Trivy is for scanning container images for security vulnerabilities.

Deliverables & Assessment

1. Evidence (label each clearly)

- The 401 (no credentials) and 200 (valid credentials) results (Task 1).
- The MFA OK output for a valid TOTP code (Task 2).
- The three auth can-i results — allowed vs denied (Task 3).
- web→db BLOCKED and app→db REACHABLE (Task 4).
- The iptables default-deny ruleset (Task 5).
- The hardened container inspect output and the Trivy scan summary (Task 6).

2. Short-Answer Questions

- Q1.** Explain the difference between authentication and authorization using Tasks 1 and 3.
- Q2.** Why is MFA so effective, and which attacks does it defeat?
- Q3.** How does network segmentation limit the damage of a compromised web server?
- Q4.** What does a **default-deny** firewall policy achieve, and how does it relate to cloud security groups?
- Q5.** List the hardening measures you applied and the attack surface each one removes.

3. Verification Command

```
kubectl get rolebinding dev-rb -n app -o yaml
docker inspect hardened --format '{{json .HostConfig.CapDrop}}'
```

Security Best-Practices Checklist

- ☐ Service requires authentication (unauthenticated requests rejected).
- ☐ MFA / second factor implemented and validated.
- ☐ Authorization enforced by RBAC (least privilege; unauthorised actions denied).
- ☐ Network segmented so the data tier is unreachable from the front tier.
- ☐ Default-deny firewall with explicit allow rules.
- ☐ Container hardened: non-root, minimal, capabilities dropped, read-only; image scanned.

Cleanup & Teardown

```
docker rm -f authsvc db app web hardened 2>/dev/null
docker network rm frontend-net backend-net 2>/dev/null
kind delete cluster --name ccse-lab4
```

Expansion Ideas (Advanced Students)

- Add a **Web Application Firewall** (ModSecurity) in front of the service and block a test SQL-injection string.
- Deploy **fail2ban** to auto-block IPs after repeated failed logins.

- Introduce a service mesh (**Istio**) and enforce **mTLS** between services (zero-trust network).
- Turn the hardening steps into a **Dockerfile** with a distroless base image and rebuild.

References

- Course lectures — Week 5 (Access Control), Week 9 (Network Security patterns).
- Docker security — docs.docker.com/engine/security
- CIS Docker / Kubernetes Benchmarks — www.cisecurity.org
- CSA Security Guidance v5 — Infrastructure & Networking; IAM.